TASK 9 REPORT

Title

Diabetes Prediction Using Machine Learning

Objective

The objective of this task is to build a machine learning model that predicts whether a person is likely to have diabetes based on medical and health-related attributes. The project aims to demonstrate the application of machine learning techniques in healthcare prediction.

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Flask
- Jupyter Notebook

Dataset Description

The Diabetes dataset contains several health-related features such as glucose level, blood pressure, BMI, insulin level, age, and other medical attributes. These features are used to predict whether a patient has diabetes.

Methodology

Step 1: Data Collection and Loading

The dataset was imported using the Pandas library and examined to understand its structure and attributes.

Step 2: Data Exploration

The following operations were performed:

- Displayed sample records
- Checked dataset dimensions
- Generated statistical summaries
- Identified missing or invalid values

Step 3: Data Preprocessing

- Handled missing values
- Removed inconsistencies
- Selected relevant features
- Split the dataset into training and testing sets
- Applied data scaling where necessary

Step 4: Model Development

A machine learning classification model was trained using the prepared dataset. The model learned patterns from the training data to predict diabetes outcomes.

Step 5: Model Evaluation

The trained model was evaluated using performance metrics such as:

- Accuracy Score
- Confusion Matrix
- Classification Report

Step 6: Model Deployment

The trained model was integrated with a Flask web application to provide a user-friendly interface for prediction.

Step 7: Prediction Testing

Various input values were tested through the web application to verify the prediction functionality.

Results

The machine learning model successfully predicted diabetes outcomes based on user-provided health information. The model achieved satisfactory performance and was integrated into a web application for practical use.

Key Findings

- Health-related attributes strongly influence diabetes prediction.
- Machine learning can effectively assist in disease prediction.
- Proper preprocessing improves model performance.
- Web deployment makes the prediction system accessible to users.

Conclusion

This project demonstrated the complete machine learning workflow, including data preprocessing, model training, evaluation, and deployment. The developed diabetes prediction system provides a simple and effective method for predicting diabetes risk.

Learning Outcomes

- Data preprocessing techniques
- Feature selection
- Machine learning model training
- Model evaluation and performance analysis
- Flask web application development
- Deployment of machine learning models